Nikhil Shetty

Principal Data Engineer — Bern, Switzerland
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Summary

Principal Data Engineer with 8+ years building batch and streaming data platforms on Databricks, Spark, Kafka and AWS. Cut a production lakehouse's latency by 82% and DBU cost by 77%, rebuilt a Databricks FinOps cost-attribution pipeline, and root-caused a 127% rise in platform spend. Also ships full-stack products: sole developer of an ERP live in production. Final-semester MSc student in Bern.

Technical Skills

Languages: Python, SQL, Scala, Java
Data: Apache Spark, Structured Streaming, Kafka, Delta Lake (CDF), Databricks, Unity Catalog, Airflow, dbt, Hive
Engineering: FastAPI, Frappe/ERPNext, Flutter, Docker, Kubernetes, Terraform, GitHub Actions, Playwright
Cloud: AWS (Glue, EMR, Athena, Redshift, Step Functions), GCP (BigQuery), Snowflake
Databases: PostgreSQL, Oracle, MS SQL Server

Experience

Tricon InfoTech Remote (US client)
Principal Data Engineer Jan 2021 – Present

- Rebuilt the Databricks FinOps cost-attribution pipeline behind the cost dashboard: system list prices with effective-dated discounts, per-user allocation of shared SQL-warehouse cost, reconciliation checks, and Solution tags across 314 job templates.
- Root-caused a 127% year-over-year rise in Databricks spend via reproducible system-table analysis: serverless SQL consumption drove 78% of growth at unchanged prices; one job drove 90.5% of a product's increase.
- Architected an event-driven Databricks Lakehouse (Kafka, Spark, Delta Lake, Medallion) processing TB-scale workloads; optimized compute, storage and pipelines to cut latency by 82% and DBU cost by 77%.
- Led a 4-member team delivering a customer-facing insights platform: Oracle and Delta Lake data into Gold-layer tables and APIs, including a 3.6 billion-row model that runs 60% faster end to end.
- Built config-driven incremental learner-analytics facts on Delta Change Data Feed and extended the shared engine with per-fact reconciliation routing; redesigned Pendo export ingestion (export-log state machine, soft deletes).
- Built Kafka Structured Streaming pipelines with late-arriving-data recovery (retries, reconciliation reports, alerts); moved orchestration from Airflow to Databricks Workflows with PagerDuty alerting.

Accenture Bengaluru, India
Data Engineer Jul 2018 – Jan 2021

- Built PySpark ETL migrating enterprise data from on-premise Oracle and Cloudera to AWS.
- Built curated datasets and dimensional models for self-service analytics on Athena and Redshift Spectrum, plus hourly and daily KPI dashboards in QuickSight.
- Stabilized business-critical reporting pipelines by monitoring platform health and resolving recurring job failures.

Projects

Fuel Station Operations Platform (Python, Frappe/ERPNext, MariaDB) In production since Oct 2026

- Sole developer of a forecourt ERP for a fuel retail outlet: shifts, nozzle meters, cash hand-overs, tanker receipts and daily stock reconciliation for 12 staff users; 59 data models, 420 automated tests, staged CI/CD.

MSc Thesis: Compact Features for Merge Conflict Prediction (Python, scikit-learn, Slurm) Ongoing

- Leakage-safe feature selection on a 969-feature multilingual dataset; reproduced the reference paper's results within 1 percentage point; experiments on the UBELIX HPC cluster.

Gym Management Platform (Flutter, FastAPI, PostgreSQL, Terraform): admin, coach and member apps with class booking and QR check-in; 338 backend tests.

Education

University of Bern, Master of Science Bern, Switzerland — 2024 – Present
BVB College of Engineering, B.Tech Information Science, CGPA 8.20/10 Hubli, India — 2014 – 2018
Awards: Best Fresh Hire of the Year; Best Impactful Engineer